

ZAYSAN, KazakhstanWMO: **366650**Lat: **47.467N**Lon: **84.917E**Elev: **1978**StdP: **13.68**Time Zone: **6.00 (E06)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-12.7	-22.4	1.7	-16.5	-18.0	2.2	-11.7	22.3	23.8	17.7	23.8	2.6	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	89.6	63.0	86.8	62.4	84.3	61.7	65.7	83.5	64.4	81.8	63.1	80.1	7.7	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	80.8	72.0	57.4	75.7	72.7	55.8	71.3	72.2	31.6	83.6	30.6	82.2	29.6	79.7	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	17.8	14.4	DB	-22.0	96.6	7.0	4.5	-27.1	99.8	-31.2	102.4	-35.1	105.0	-40.2	108.2	(4)
(5)			WB	-21.9	67.9	6.7	1.2	-26.7	68.7	-30.6	69.4	-34.4	70.1	-39.3	70.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	5.7	5.9	21.9	45.6	59.8	69.7	74.7	71.1	59.6	44.7	23.5	12.4		
	DBStd	26.85	9.56	10.39	12.17	10.39	8.58	7.16	5.55	6.48	8.26	9.97	13.41	10.63		
	HDD50	5907	1374	1236	873	203	22	0	0	0	17	223	794	1165		
	HDD65	9377	1839	1656	1336	583	202	34	5	17	199	632	1244	1630		
	CDD50	2770	0	0	3	69	326	592	764	653	305	57	1	0		
	CDD65	766	0	0	0	0	41	176	304	206	38	1	0	0		
	CDH74	5936	0	0	0	2	283	1343	2430	1626	244	8	0	0		
	CDH80	1784	0	0	0	0	53	393	774	521	42	1	0	0		
(14) Wind	WSAvg	5.6	2.9	3.1	3.4	7.0	8.3	7.8	7.4	7.0	6.6	5.4	4.3	3.5		
(15) Precipitation	PrecAvg	11.9	0.4	0.4	0.6	1.1	1.6	1.2	1.7	0.9	1.0	1.2	1.0	0.7		
	PrecMax	19.0	0.9	0.9	1.5	3.1	4.6	3.0	3.8	2.6	3.2	3.5	2.5	1.8		
	PrecMin	5.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2		
	PrecStd	3.0	0.3	0.2	0.4	0.7	1.1	0.7	1.1	0.7	0.6	0.8	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	31.3	60.9	72.7	85.8	91.4	94.2	93.6	85.0	73.8	56.2	35.9		
		MCWB	27.4	28.1	48.6	53.2	62.1	63.6	65.1	64.0	59.2	55.3	46.7	32.3		
	2%	DB	24.0	26.7	49.6	69.8	80.7	87.2	89.8	89.3	80.1	67.7	47.5	32.4		
		MCWB	22.1	23.7	41.9	52.3	59.3	62.0	63.7	62.0	58.0	51.3	42.0	30.0		
	5%	DB	20.7	23.8	43.2	65.5	76.9	84.7	87.0	85.5	76.6	63.3	41.8	29.8		
		MCWB	19.1	21.4	37.3	50.1	57.7	61.4	63.2	61.3	56.7	49.1	36.4	27.9		
	10%	DB	18.2	20.7	37.1	61.3	73.6	81.7	84.4	82.4	72.7	58.8	37.8	26.7		
		MCWB	16.9	18.8	33.0	48.4	56.4	60.6	62.8	60.6	55.1	46.8	34.0	25.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.8	29.0	48.6	55.8	64.4	66.1	67.4	66.4	60.7	56.1	47.4	33.3		
		MCDB	30.1	31.2	59.2	69.0	82.5	86.5	83.6	86.5	80.2	71.8	54.4	35.0		
	2%	WB	22.3	24.2	42.6	53.0	61.7	63.8	66.3	64.1	59.0	53.0	42.6	30.1		
		MCDB	23.6	26.6	49.2	67.3	76.4	82.0	83.5	82.6	77.1	65.6	48.0	32.0		
	5%	WB	19.2	21.4	37.5	51.1	59.4	62.6	65.2	62.7	57.5	49.7	37.1	28.0		
		MCDB	20.4	23.6	43.0	64.5	73.8	79.8	82.3	81.1	73.3	61.9	41.2	29.8		
	10%	WB	16.8	18.9	33.0	48.6	57.5	61.5	64.0	61.5	56.1	47.4	34.2	25.0		
		MCDB	18.1	20.8	36.9	60.9	71.3	78.9	80.8	79.5	71.2	58.1	37.7	26.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	12.7	12.3	16.3	16.4	15.6	14.5	15.2	16.9	15.3	11.7	10.6		
		MCDBR	14.0	14.2	16.7	21.9	21.3	18.2	17.5	18.8	21.8	20.0	17.0	12.2		
	5% WB	MCWBR	12.3	12.2	11.4	12.4	11.7	8.0	7.1	7.9	10.4	11.1	12.1	10.3		
		MCWBR	13.8	14.2	16.6	20.2	18.0	16.2	15.0	16.9	19.1	18.8	16.2	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.327	0.379	0.427	0.411	0.417	0.420	0.392	0.358	0.351	0.325	0.295		
	taud		2.138	2.097	2.064	2.057	2.167	2.230	2.245	2.300	2.342	2.255	2.189	2.134		
	Ebn at Noon		241	258	262	261	271	269	266	269	248	226	226	223		
	Edh at Noon		28	36	44	49	45	43	42	38	33	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		569	907	1307	1642	2055	2121	2029	1876	1469	896	512	423		
	RadStd		29	40	89	111	113	120	155	95	100	84	44	33		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page